Build guide 1: tools and steel

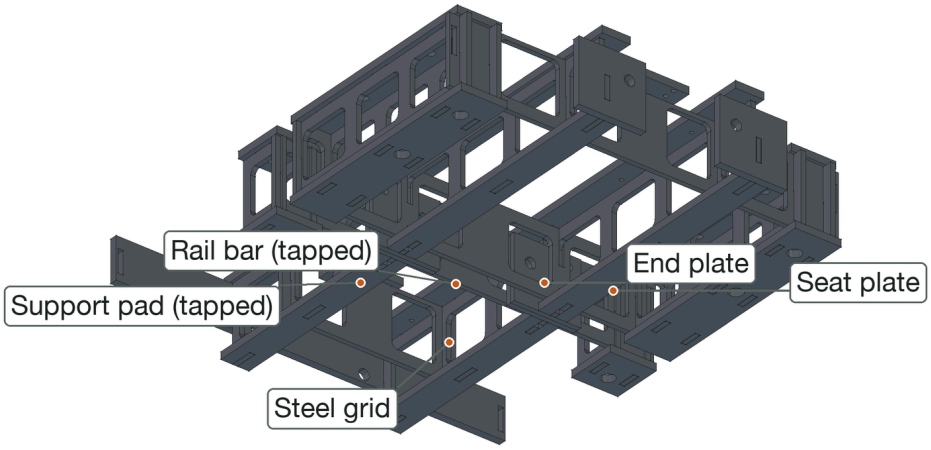
DIY CNC epoxy-granite mill · October 2026

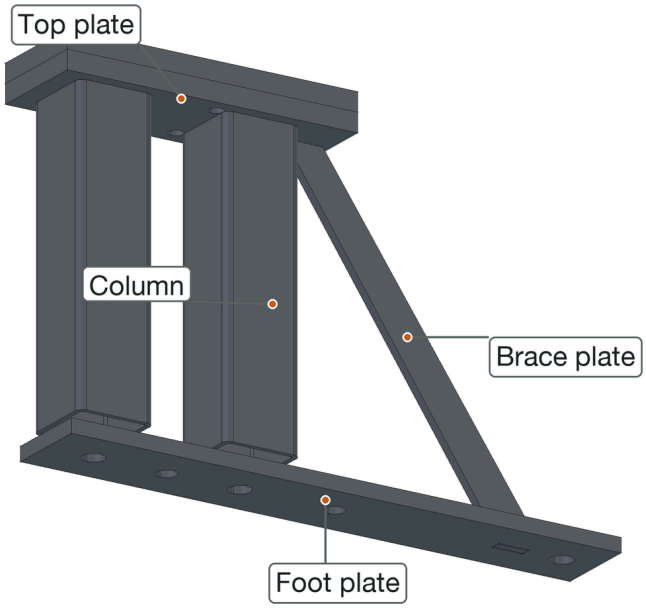
0. Tools, space and safety

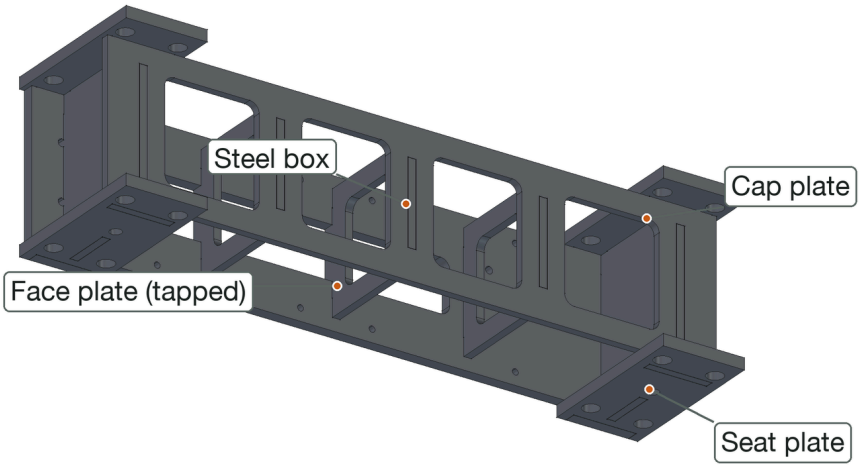
- **Space:** a level bench or floor area of about 1.2 x 1.5 m at 20-25 °C for a week. Epoxy cures slowly and badly below 18 °C.
- **Tools:** drill press, chop saw or hacksaw, angle grinder with flap disc, taps (M4, M5, M6), drill/mixer with a paddle (or a small cement mixer), kitchen scale + a 50 kg scale, vibration source (a table vibrator, a concrete poker, or a reciprocating sander with the pad removed pressed against the mold), straightedge (at least 600 mm, ideally ground), feeler gauges, dial indicator with a magnetic base, precision level, torque wrench.
- **Safety:** nitrile gloves and long sleeves for epoxy (skin sensitisation is permanent); an N95 dust mask when pouring the whiting; a P100 respirator when grinding or sanding cured EG (the quartz sand and gravel in it make silica dust; work wet if you can); eye protection when cutting steel. Mix in a ventilated space.
- **3D printing (PETG):** the mold kit (brackets, saddles, holders, bushings), the core posts, formers and fillers, the parts that get cast in (column rings, tube plugs, drain liner) and the machine's printed parts. All of it is in `print/`, split to fit a Bambu A1 mini, and listed in the print guide: 1.1 kg of PETG (sliced), inside two 1 kg rolls, and about 90 hours of printing. The base pocket cores are cut from one sheet of 2 in XPS foam board instead. Print one bracket and one saddle first and check them on your board; print a column ring early and test-fit it in your actual tube.
- **Foam cutting:** a fine-tooth saw or a hot-wire cutter, a long knife, a sanding block, packing tape and double-sided carpet tape.

1. Buy and prepare the steel

The frame's backbone is steel: a skeleton in each casting, slotted together from laser-cut plates and clamped with threaded rod, then cast into epoxy granite, which damps it and stiffens it further. Every joint between the castings, and everything that bolts on (rails, ballscrew supports), is steel to steel. Nothing is welded and nothing is threaded into the epoxy granite.

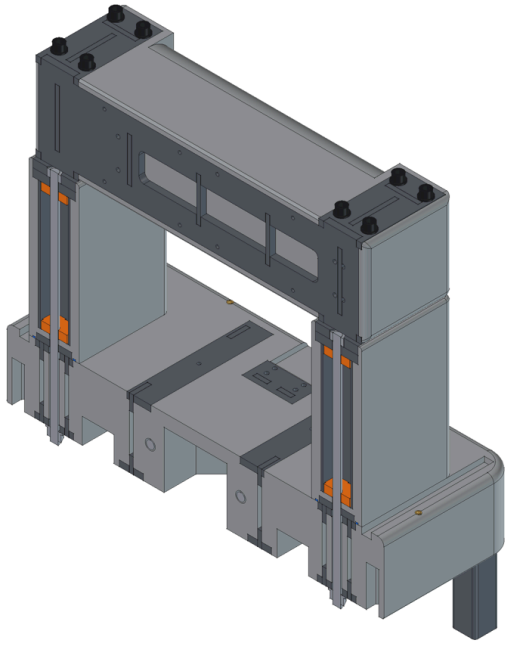

Rail bar (tapped)
Support pad (tapped)
End plate
Seat plate
Steel grid


Top plate
Column
Brace plate
Foot plate



1. **Laser-cut plates.** Send the DXFs in `docs/1-sourcing/laser/` to Metal Supermarkets, a laser-cutting service, or a friend with a CNC plasma or router table. The README there lists every part, its thickness and how many. About 64 parts, roughly 48 kg of 1/4, 3/8 and 1/2 in mild steel. The tabs on one plate drop into slots in the next, so each skeleton only goes together one way.
   - Knock the dross off every edge and slot; the tabs should slide in by hand.
   - **Tap** the holes the README lists, at the drill size they're cut to: 3/8-16 in the upright top plates' lower layers, M5 in the base rail bars and the beam face plate, M6 in the base support pads and the beam face plate. Use a tap guide or the drill press to start them square.
2. **Square tube** (plain steel, not galvanized; 0.120 or 1/8 in wall; remnants are fine), bought at length: 2 x 2 in, 4 x 8 in (the upright columns); 1-1/2 x 1-1/2 in, 4 x 6 in (legs).
   - A saw cut within 1/8 in is fine: a column is clamped up against its top plate and its foot stands in an epoxy bed on the foot plate (4.8 mm nominal); a leg is pushed up against its leg plate and its leveling foot takes up the rest. Knock off burrs.
   - The printed rings that center a column on its rod have 0.5 mm a side and grooves for the weld seam, so they fit tube up to 0.5 mm off. If yours is further off, put its calipered sizes in `frame_tube` / `post_tube` in `model/build_cnc.py`, re-run it and print the rings again.
3. **Slot each skeleton together on the bench** (pictures above, and in the mold guide):
   - **Upright** (two): brace into the foot plate; columns on the foot plate, a printed ring in each end; top plate's lower layer over the brace tab, upper layer on it. It's clamped for good later, by the tie rods.
   - **Beam:** diaphragms into the face plate's slots, back plate over the diaphragms' tabs, the seat and cap plates onto the end tabs.
   - **Base** (one grid per half): girder webs into their bottom bars and end plates, pedestal webs into their bars, the cross plates dropped over the webs (half-depth slots, so they lock), then the rail bars, seats and pads on top and the leg plates through their cross plates. Glue the four 3/8 x 1 in dowel pins into the rear half's split end plates, half their length out.

4. **Rail risers** (cold-rolled 1018, 1-1/4 in wide, ordered cut to the rail lengths): Y 3/4 in x 600 mm (2), X 5/8 in x 400 mm (2), Z 5/8 in x 300 mm (2). The rails are used at full length.
    - Clamp each rail on its riser, centered, and use the rail's own holes as the drill template; drill and tap M4 (60 mm pitch). Mark which rail goes on which riser.
    - Y and X risers also get the hold-down holes, midway between rail holes and under the rail: 5.5 mm through with a 9.5 mm x 4.5 mm deep counterbore from the top. Positions from the riser center: Y at 0, ±120 and ±240 mm, X at ±30 and ±150 mm. They go into the tapped rail bars (base) and face plate (beam).
    - Stone or lightly sand off burrs; keep the top faces flat.
5. **Clean all the steel:** wire-brush or flap-disc to dull gray metal, then degrease with acetone just before casting. If it will sit more than a day or two, give it a thin coat of epoxy primer (not on the faces the mold screws or the rails bear on). Paint only the part of each leg that will hang below the base (after casting).
6. **Threaded rod and hardware:** cut the 1/2-13 rods: per upright 2 x 375 mm (in the columns) and 1 x 153 mm (at the toe); 2 x 687 mm for the base draw rods. Deburr the ends. A rod a few mm off is fine: its top end is set flush and the error ends up in the 6 mm tail past the bottom nut. Have 11 x M5 cast-in inserts ready for the enclosure (the only inserts left).



The cast-in steel for each base half (castings drawn see-through):

